- Using either event streams with a Kafka topic carrier as data output ports, or directly mapping the topic into a provided output format like a Snowflake table or a CSV file inside cloud storage using Kafka Connect sinks.
- Defaulting to ksqlDB as SQL-syntax-based transformation tooling inside the data product. Additionally, ksqlDB is used as a tool for ingesting data from event stream data output ports that are carried by a Kafka topic.

Explanation of Kafka components

The following is a short description of the Kafka components used in this section:

- *Apache Kafka*—An open source streaming platform for high-performance streaming and analytics. Data inside Kafka is organized into topics. The default way of getting data into Kafka is to use the Producer API, e.g., inside Java. Similarly, the Consumer API can be used to get data out of Kafka.
- *Kafka Connect*—For diverse data sources, Connect offers a variety of connectors to import data into topics as with any Java Database Connectivity (JDBC) source, as well as export data (called sinks) into other data systems (for instance, typical object storage systems).
- *ksqlDB*—This distributed open source streaming database, built on top of Kafka, allows transforming data inside Kafka by using a SQL syntax.

You should be able to quickly figure out a few important highlights in this architecture:

- A lot is going on inside the Kafka ecosystem.
- The data inputs are standardized using Kafka sources, which is something that wasn’t the case in most other setups.
- The data output ports are standardized using Kafka sinks and the event streams carried by Kafka topics.
- KSQL is used for transformations inside the ecosystem.